

TSA Home Assignment 1

As announced earlier, grading for HWs consists of points and bonus points. Solving bonus (indicated with a star) problems is not required, but recommended. Solving all homeworks' normal problems correctly will give you a score of 7, solving all homeworks' bonus problems correctly will give you additional 2 points to the score.

Hand-written solutions are accepted if the handwriting is clear enough and scanned with sufficient quality, but LaTeX is always preferable. This homework includes a python task, which can be solved in Google Colab or in a local Jupyter Notebook. It is thus handy to solve everything (both LaTeX and code) in a single Jupyter Notebook.

Solutions obtained with the use of ChatGPT and similar models can be accepted if the solution is clearly indicated as such, and model version and prompt is provided. If the solution is found to be from ChatGPT and similar models without indication and model/prompt detail, the teachers will evaluate the problem as zero points.

Problem 1. Consider AR(2) process

$$X_t = X_{t-1} - \frac{1}{2}X_{t-2} + Z_t$$

1. Find the characteristic polynomial $\varphi(B)$.
2. Write out equation $\varphi(B) = 0$, find its roots.
3. Locate roots in or out complex unit circle, make conclusion about stationarity.
4. Write out concrete Yule-Walker equation for $k = 1$, use it to find ACF at lag 1, $\rho(1)$.
5. Write out Yule-Walker equation for $k > 1$, its general form solution for $\rho(k)$.
6. Write out auxiliary equation, find its roots, make conclusion about form of ACF $\rho(k)$ (gradual decay vs damped sinusoid).

Problem 2. Consider ARMA(1, 1) process

$$X_t = 0.5X_{t-1} + Z_t - 0.3Z_{t-1}$$

1. Find the characteristic polynomial $\varphi(B)$.
2. Find the characteristic polynomial $\theta(B)$.
3. Find the characteristic polynomial $\psi(B) = \theta(B)/\varphi(B)$.
4. Find the characteristic polynomial $\pi(B) = \varphi(B)/\theta(B)$.
5. Make conclusions about stationarity.

Problem 3. For ARIMA model

$$(1 - B)(1 - 0.2B)X_t = (1 - 0.5B)Z_t$$

1. Identify p, d, q
2. Determine if this process is stationary, if it is invertible.
3. Find first three coefficients in $\psi(B) = \theta(B)/\varphi(B)$.
4. Find first four coefficients in $\pi(B) = \varphi(B)/\theta(B)$.